Static Black Holes

The Schwarzschild Solution

We start with the Schwarzschild solution: The uniqueness of this sol^{n} is determined by Birkhoff's theorem.

$$ds^2 = -\left(1 - \frac{2Gm}{r}\right)dt^2 + \left(1 - \frac{2Gm}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2\theta \, d\phi^2)$$

We will think of m as the mass of the black hole

Recall that the killing vector k is the field corresponding to the flow $\sigma : M \to M$. This creates an isometry and allows for conservation of quantities. Schwarzschild spacetime admits a timelike killing vector $k = \partial_t$.

The dual one-form for this will be given by: $k = g_{00} dt = -\left(1 - \frac{2Gm}{r}\right) dt$

We are trying to make this like a Komar integral, so:

$$F = dk = -\frac{2Gm}{r^2} dr \wedge dt \longrightarrow M_{Komar} = -\frac{1}{8\pi G} \int_{S^2} *F$$

↳ Notice similarities to E. field ↳ S^2 is any sphere w $r > 2Gm$

S^2 is analogous to the Gaussian surface, any r encapsulates the same.

Now, we know that $F = dk$ obeys $d*F = 0$, analogous to the vacuum Maxwell eq^{n}s in the absence of current. This means that the electric charge must vanish. Yet it doesn't happen.

What this tells us that for the black hole, the mass is all localised at the origin where the field strength diverges.

As an aside, in GR, time independence can have two different meanings:

① Stationary → A timelike killing vector is admitted everywhere.

② Static → Stationary + invariant under $t \to -t$. This rules out $dt\,dx$ cross terms in the metric.

Horizons of Schwarzschild Space

The surfaces $r = 0$ and $r = 2Gm$ have divergences. The divergence at the $r = 0$ surface is a feature of the classical theory and is called the **singularity**.

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The divergence at $r = 2GM$ is due to a bad choice of coordinates. This surface is called the Event Horizon.

A litmus test for understanding if the divergence is inherent is to construct a coordinate independent scalar and check its singularity. The simplest curvature scalar for our case is the Kretschmann Scalar $R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}$ which, for our solution is:

$$R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6}$$ clearly gives no divergences at the event Horizon.

Note also that heavier black holes have smaller curvature at the horizon.

If we want to scope out the event Horizon of the blackhole, we can zoom in and look at the metric near the horizon:

$r = 2GM + \eta$ → $\eta \ll 2GM$, $\eta > 0$ so looking just outside.

Approximating the metric:

$$ds^2 = -\frac{\eta}{2GM}dt^2 + \frac{2GM}{\eta}d\eta^2 + (2GM)^2 d\Omega_2^2$$

The metric is a direct product of an S^2 of radius $2GM$ and a $d = 1+1$ dimensional Lorentzian geometry. We shift our focus to the Lorentzian geometry.

$$\rho^2 = 8GM\eta \rightarrow ds^2 = -\left(\frac{\rho}{4GM}\right)^2 dt^2 + d\rho^2$$

Keep in mind ρ is a rescaled version of $\sqrt{r}$

The above metric is known as Rindler space. If we then transform:

$$T = \rho\sinh\left(\frac{t}{4GM}\right), \quad X = \rho\cosh\left(\frac{t}{4GM}\right) \rightarrow ds^2 = -dT^2 + dX^2$$

→ Proper time.

The metric under these coordinates ends up resembling Minkowski space. What does this mean physically? The coordinates X and T are those of an observer undergoing a constant acceleration $a = 1/4GM$. Physically then, its saying that if you want to sit at a constant ρ naturally, you need to keep accelerating.

We can now use these coordinates to map out the bit of Minkowski space outside the blackhole horizon: $\rho > 0$, $t \in (-\infty, +\infty]$ → $X > |T|$

As for the horizon itself, its a bit trickier.

For any finite t, horizon $\rho = 0$ is mapped to the origin of Minkowski space. But $g_{00} = 0$ so we scale $t \to \infty$, $\rho \to 0$ keeping $\rho e^{\pm t/4am}$ fixed to find that the horizon corresponds to:

$r = 2am \to x = \pm T$ The Horizon is a Null surface!

This analysis also makes it clear that there is nothing inherently wrong about spacetime at the event horizon!